

PAPER 1 / QUESTION 64

29 April 2025 / TZ1 (file label) / HL Paper 1A / original question 4 / 2225-6707

- 64.** A sphere of density ρ and radius R rests on the bottom of a tank of water. The buoyancy force on the sphere is F_1 .

A sphere of density 2ρ and radius $\frac{R}{2}$ is at the bottom of the same tank. The buoyancy force on the second sphere is F_2 .

What is $\frac{F_1}{F_2}$?

- A. 2
- B. 4
- C. 8
- D. 16